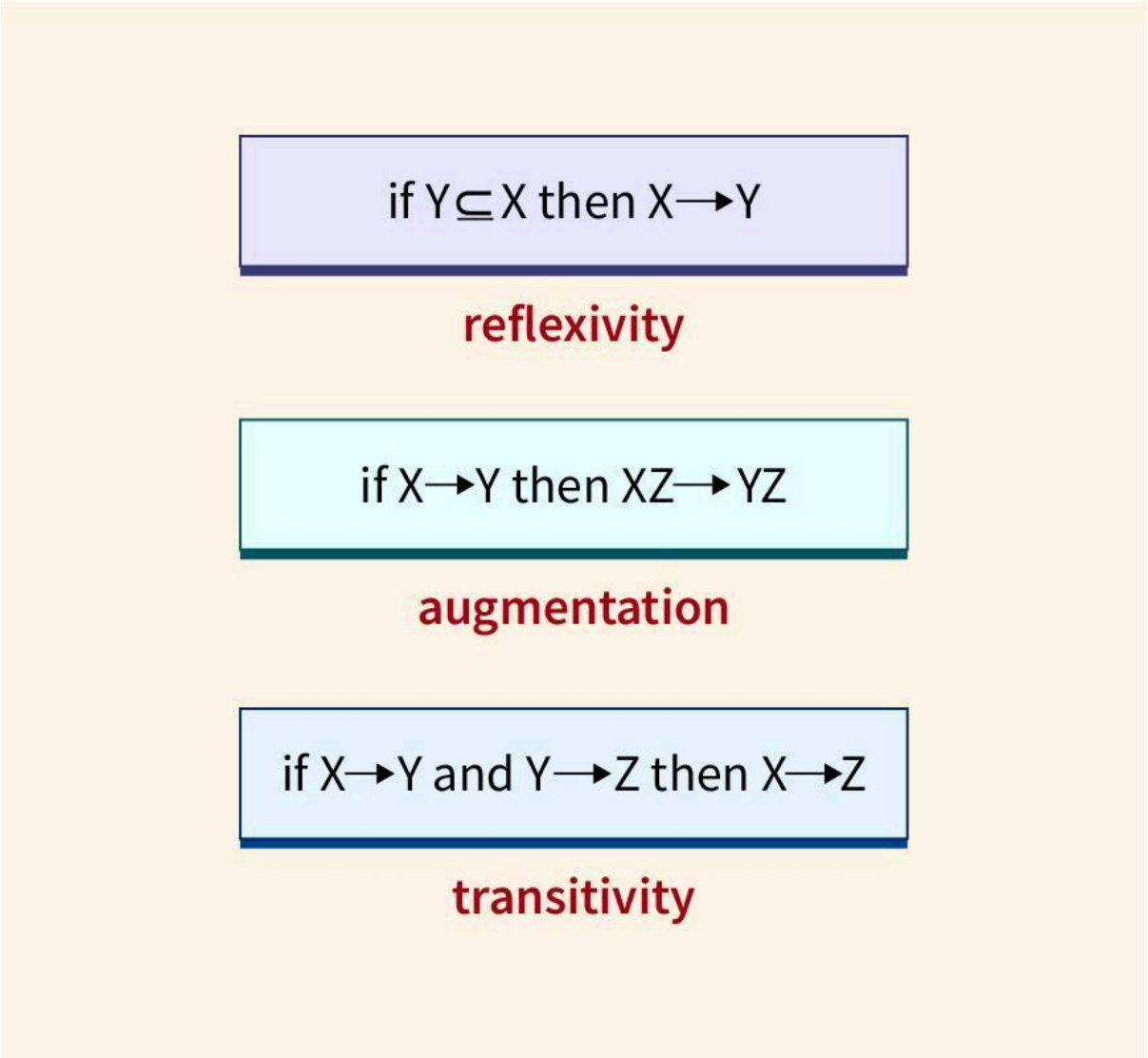
**Definition:**
Armstrong's Axioms are a set of **inference rules** used to derive **new functional dependencies (FDs)** from a given set of functional dependencies.

They were proposed by **William W. Armstrong**.

🔑 **Three Basic Axioms**



# Armstrong's axioms

- **Reflexivity**: If $Y \subseteq X$ then $F \setminus X \rightarrow Y$
  - (This is called a trivial dependency)
  - Example: `sname,address→address`
- **Augmentation**: If $F \setminus X \rightarrow Y$ then $F \setminus X,W \rightarrow Y,W$
  - Example: As `cid→cname` then `cid,sid→cname,sid`
- **Transitivity**: If $F \setminus X \rightarrow Y$ and $F \setminus Y \rightarrow Z$ then $F \setminus X \rightarrow Z$
  - Example: As `sid,cid→cid` and `cid→cname`, then `sid,cid→cname`



## Armstrong Axioms

All implied FDs can be derived using the Armstrong axioms.

### Armstrong axioms

- **Reflexivity:** if $\beta \subseteq \alpha$, then $\alpha \rightarrow \beta$.
- **Augmentation:** if $\alpha \rightarrow \beta$, then $\alpha \cup \gamma \rightarrow \beta \cup \gamma$.
- **Transitivity:** if $\alpha \rightarrow \beta$ and $\beta \rightarrow \gamma$, then $\alpha \rightarrow \gamma$.

Use the Amstrong axioms to show that

1. `isbn → title,publisher`
2. `isbn,no → author`
3. `publisher → publisherURL`

implies `isbn → publisherURL`.

4. `title,publisher → publisher` *by reflexivity*
5. `isbn → publisher` *by transitivity using 1. and 4.*
6. `isbn → publisherURL` *by transitivity using 5. and 3.*

### 1️⃣ Reflexivity Rule

If **Y ⊆ X**, then:

**X ⟶ Y**

📌 **Example:**
{Student_ID, Name} ⟶ Student_ID

Since **Student_ID** is a subset of {Student_ID, Name}, the dependency is valid.

**Remember:**
👉 *A set of attributes determines any subset of itself.*

---

### 2️⃣ Augmentation Rule

If:

**X ⟶ Y**

then:

**XZ ⟶ YZ**

📌 **Example:**

Given:

**Student_ID ⟶ Name**

Add the same attribute **Course_ID** to both sides:

**Student_ID, Course_ID ⟶ Name, Course_ID**

**Remember:**
👉 *Add the same attribute(s) to both sides.*

---

### 3️⃣ Transitivity Rule

If:

**X ⟶ Y**
and
**Y ⟶ Z**

then:

**X ⟶ Z**

📌 **Example:**

Student_ID ⟶ Department_ID
Department_ID ⟶ Department_Name

Therefore:

**Student_ID → Department_Name**

**Remember:**
👉 *If X determines Y and Y determines Z, then X determines Z.*

---

## ⭐ Derived Rules

From the three basic axioms, additional rules can be derived.

| Rule | Name | Formula |
|---|---|---|
| 1 | Union | If X → Y and X → Z, then **X → YZ** |
| 2 | Decomposition | If X → YZ, then **X → Y** and **X → Z** |
| 3 | Pseudotransitivity | If X → Y and WY → Z, then **WX → Z** |

## 🧠 Easy Memory Trick

**R – A – T**

**R**eflexivity → **A**ugmentation → **T**ransitivity

# Armstrong's Axioms

**Definition :**

Armstrong's Axioms are a set of inference rules used to derive new functional dependencies (FDs) from a given set of functional dependencies.

They were proposed by **William W. Armstrong**.

## Three Basic Axioms :

### ① Reflexivity Rule :

If $Y \subseteq X$, then $X \rightarrow Y$

Example : {Student_ID, Name} → Student_ID

Since Student is a subset of {Student_ID, Name}, the dependency is valid.

**Remember :**
A set of attributes determines any subset of itself.

### ② Augmentation Rule :

If $X \rightarrow Y$, then $XZ \rightarrow YZ$

Example : Given Student_ID → Name

Add the same attribute Course_ID to both sides.

Student_ID, Course_ID → Name, Course_ID

**Remember :**
Add the same attribute(s) to both sides.

### ③ Transitivity Rule :

If $X \rightarrow Y$, and $Y \rightarrow Z$, then $X \rightarrow Z$

Example : Student_ID → Department_ID

Department_ID → Department_Name

Therefore, Student_ID → Department_Name

**Remember :**
If X determines Y and Y determines Z, then X determines Z.

## Derived Rules :

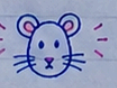
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## Easy Memory Trick :

R – A – T

"RAT"



R → Reflexivity
A → Augmentation
T → Transitivity

These three are the fundamental Armstrong's axioms used to infer functional dependencies.